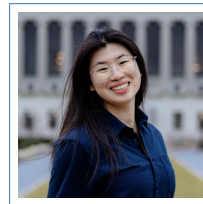


ANNA CHEN

Curriculum Vitae

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Legal Name: Yiyu Chen (陈一语)

Work Authorization: U.S. Permanent Resident

Fellowship Applications: NSF Graduate Research Fellowship Program (GRFP) - In preparation for Fall 2026 submission

Education

2023–2027 **B.A. in Climate System Science & B.A. in Applied Mathematics**, *Columbia College, Columbia University*, New York, USA.

GPA : 4.02/4.00

2023 **Higher Secondary Examination: International Baccalaureate Bilingual Diploma**, *YK Pao School*, Shanghai, China.

Score : 44/45

Research Experience

[Lamont-Doherty Earth Observatory \(LDEO\)](#), [Columbia University](#)

Spring 2026 – Present [Senior Thesis] **Evaluating the Impacts of Model Choice on Assessment of Surface Ocean pCO₂ Reconstruction Skill in Climate Model Testbeds.**

2-credit independent research course + 2026 Comer Climate Summer Research Fellowship + Senior Thesis

Mentor : **Dr. Galen McKinley**, *Professor, Dept. of Earth and Environmental Sciences and LDEO*, Columbia University ([Lab web-page](#))

Spring 2025 – Present **Global Reconstruction of Last Interglacial Sea Surface Temperatures and Their Uncertainties by an Iterated Reduced-Space Optimal Interpolation Procedure.**

Data Science Institute Scholarship; Heinrich Family Global Fellowship

Interpolated, visualized, and animated global-field sea-surface temperature from sparse paleo-proxy observations over the Last Interglacial using Reduced-Space Optimal Interpolation; coding in Python.

Contributed to NSF-funded research: [HISEAS](#)

PI/Mentor : **Dr. Jerry McManus**, *Professor, Dept. of Earth and Environmental Sciences; Deputy Director for Education, LDEO, Earth and Environmental Sciences*, Columbia University

Mentor : **Dr. Alexey Kaplan**, *Associate Research Professor, LDEO*, Columbia University

Summer 2024 – Spring 2025 **Seasonal and Geospatial Analysis on Geochemical Ocean Conditions in Areas with High Marine Phytoplankton Productivity.**

Laidlaw Research and Leadership Fellowship

Visualized and analyzed seasonal and geospatial patterns of satellite-derived marine net primary production based on sea-surface temperature and pCO₂; coding in R-language.

Practiced phytoplankton culturing, flow-cytometry, media making, and sterile lab methods

PI/Mentor : **Dr. Sarah Hurley**, *Assistant Research Professor, LDEO*, Columbia University ([Lab web-page](#))

Mentor : **Dr. Henry Holm**, *Postdoctoral Research Scientist, LDEO*, Columbia University

Selected Final Projects for Courses at Columbia University

Fall 2025 **Baroclinic Modes and Frontal Variability of Zonal Flow in the Antarctic Circumpolar Current: Evidence From the Drake Passage Using Singular Value Decomposition**, APMA E4300 Computational Math: Intro to Numerical Methods.
Programming in Python

Mentor: **Dr. Andreas Thurnherr**, *Research Professor, LDEO; Adjunct Professor, Dept. of Earth and Environmental Sciences, Columbia University*

Fall 2025 **Elevation-Dependent Surface Mass Balance Controls on Ice-Divide Dynamics: Iterative Shallow-Ice Modeling with Antarctic Temperature Stacks**, EESC GU4220 Glaciology.
Programming in Python

Manuscripts in Preparation

Chen, A., McKinley, G. A., Heimdal, T. H., Fay, A. R. (2027). Testbed Experiments for Surface Ocean CO₂ Mapping: Machine Learning Approaches. *In progress, Senior Thesis at the Department of Earth and Environmental Sciences, Columbia University.*

Heimdal, T. H., Ford, D. J., Keppens, M., Rogerson, J., Barton, A., Chevallier, F., Roobaert, A., Fay, A. R., McKinley, G. A., Gehlen, M., **Chen, A.**, Shutler, J. D., & Landschützer, P. (2026). The SOCOMv2 initiative: comparison of surface ocean pCO₂ data products using the CMIP6-testbed. *Manuscript in preparation for submission.*

Holm, H. C., Schaum, A. P., Ellis, A., Acosta, B., **Chen, A.**, Wing, B., McKinley, G. A., Dyhrman, S. T., & Hurley, S. J. (2026). Using Climate Models to Inform Multi-Driver Experiments with Future Ocean Conditions: Applications to Phytoplankton. *Manuscript in preparation for submission to Global Change Biology.*

Fellowships & Awards

Summer 2026 **Comer Climate Undergraduate Summer Research Fellowship**, Columbia University, (\$9,600).
Funds 4 Columbia College students to participate in a 10-week supervised summer research related to climate, sustainability, and environmental justice.

Summer 2025 **Columbia College Summer Research Fellowship**, *Heinrich Family Global Fellowship*, (\$3,600).
Awarded to non-seniors at Columbia College for academic excellence to pursue 6-weeks of independent summer research with faculty mentorship

Spring 2025 **Data Science Institute Scholars Program**, Columbia University, (\$3,000).
Awarded to mostly Master's, and some highly achieving Bachelor's students, to pursue mentored research in a data-science-related field during an academic semester (15 weeks)

2024-2025 **Laidlaw Undergraduate Research and Leadership Scholarship**, Columbia University, (\$7,600 + housing).
Funds Columbia College first-years who wish to pursue summer research, community engagement, and leadership across all disciplines. Full-time stipend awarded across two summers, 6-weeks in each

All Semesters **Dean's List**, Columbia College, Columbia University.
Awarded to students with GPA 3.6 and above

2023 **Valedictorian**, *YK Pao School*, Shanghai, China.
Top academic performance, received \$28,000 in merit-based scholarship. Voted as one of two valedictorian speakers on the High School Class of 2023 graduation ceremony

Teaching Experience

- Spring 2027 **EESC GR4926: Intro to Chemical Oceanography**, Columbia University.
Teaching Assistant
- Fall 2025 **EESC UN2100: Earth's Environmental Systems: Climate**, Columbia University.
Teaching Assistant
- Summer 2025 **STEM Mentoring**, Roaring Cubs Collective.
Mentored 3 NY public high-schoolers on writing literature review in climate and environmental sciences
Hosted weekly 1-on-1 office hours, edited student publications, paneled coaching sessions for 100+ students

Presentations

- Fall 2026 **AGU 2026 Annual Meeting**, San Francisco.
Poster 1: "Evaluating the Impacts of Model Choice on Assessment of Surface Ocean pCO₂ Reconstruction Skill in Climate Model Testbeds"
Session: A072 - From Physics to AI: Benchmarking, Bias Diagnosis, and Process Representation in Earth System Models
- Poster 2:** "Global Reconstruction of Last Interglacial Sea Surface Temperatures and Their Uncertainties by an Iterated Reduced-Space Optimal Interpolation Procedure"
Session: PP008 - Late Quaternary Climate and Environmental Change: Proxies, Processes, and Models
- Fall 2026 **Undergraduate Research Symposium**, Columbia University.
Poster: Evaluating the Impacts of Model Choice on Assessment of Surface Ocean pCO₂ Reconstruction Skill in Climate Model Testbeds
- Summer 2026 **McKinley/Lovenduski Group Meeting**, University of Colorado-Boulder.
Talk: "Evaluating the Impacts of Testbed Model Choice to Global Ocean Surface pCO₂ Reconstruction"
- Spring 2026 **Mini-Symposium on AI-based Science**, LDEO.
Talk: "Can Selecting Testbed (CMIP6) Models Based on the Representation of Seasonality Improve Global Surface-Ocean pCO₂ Modeling?"
- Fall 2025 **Geochemistry Poster Session**, LDEO.
Poster: "Last Interglacial Sea-Surface Temperature Reconstruction: A Preliminary Approach Using Reduced-Space Optimal Interpolation"
- Fall 2025 **North America Laidlaw Scholars Conference**, Brown University.
Talk: "Farming Futures: Building Colorado Communities Through Environmental Stewardship"
- Spring 2025 **Scholars Program Poster Presentation**, *Data Science Institute*, Columbia University.
Poster: "Visualizing Sparse Sea Surface Temperature Data From the Last Interglacial"
- Fall 2024 **North America Laidlaw Scholars Conference**, Columbia University.
Panel/Talk: "Seasonal and Geospatial Analysis on Geochemical Ocean Conditions in Areas with High Marine Phytoplankton Productivity"

- Fall 2024 **LDEO 75th Anniversary Mini Symposium**, LDEO.
Poster: “Seasonal and Geospatial Analysis on Geochemical Ocean Conditions in Areas with High Marine Phytoplankton Productivity”
- Fall 2024 **Undergraduate Computer and Data Science Research Fair**, *Data Science Institute*, Columbia University.
 Same poster as above
- Fall 2024 **Undergraduate Research Symposium**, Columbia University.
 Same poster as above

Leadership and Volunteering

- Fall 2024 – **President**, *Marie Tharp Society*, Dept. of Earth and Environmental Sciences, Columbia University.
 Present Lead the official undergraduate student society of the Dept. of Earth and Environmental Science (DEES) and Lamont (LDEO)
 Host events: social mixers, faculty Q&As and panels, geology movie nights, and student-led Lamont Tours
- Spring 2024 – **President**, *Zen Buddhism Club*, Columbia University.
 Present Organize weekly in-person retreats at Columbia, and bi-monthly meditation at Columbia and NYU with 130 attendees
 Book event venues, coordinate campus access, maintain social media & PR, represent club at campus club fairs
- Fall 2025 – **Vice President**, *Columbia University Cycling Team*.
 Present Coordinate training (4/wk) and racing, apply for school recognition, recruit members, oversee WhatsApp group chat
- Fall 2025 – **Volunteer**, *Climate Lab*, American Museum of Natural History.
 Present Explain climate processes to museum visitors, typically children, through hands-on activities, e.g. the thermohaline circulation, the biological carbon pump, solar radiation & albedo, the greenhouse effect
- Fall 2024 & 2026 **Volunteer**, *Open House*, LDEO.
 Educated children on paleo-botany through making bookmarks from plant-extract paper chromatography
- Summer 2025 **Volunteer**, *Fleischer Family Farm*, Colorado.
 Volunteered as [WWOOFer](#) on a 2-acre organic farm near Denver, CO for 6 weeks as a Laidlaw Scholar
 Practiced weeding, soil tilting, transplanting seedlings, harvesting, biological pest control; hosted weekly farm-stands
- Summer 2024 **Volunteer**, *World Oceans Day*, United Nations.
 Facilitated guest check-in and backstage set-up for event with 30+ speakers and 350+ attendees at UN Headquarters

Relevant Courses

Earth and Environmental Sciences

- EESC UN2100 Earth’s Environmental Systems: Climate
- EESC UN2330 Science for Sustainable Development
- EESC UN3000 Tutorial in Earth Science
- EESC UN3031 Chemistry of Climate
- EESC UN3400 Computational Earth Science
- EESC UN3901 Senior Seminar
- EESC GR4220 Glaciology
- EESC GR4926 Intro to Chemical Oceanography

- EESC GR4937 Cenozoic Paleooceanography
- EESC GR4925 Intro to Physical Oceanography

Chemistry

- CHEM UN1404 General Chemistry II
- CHEM UN3079 Physical Chemistry I: Thermodynamics
- EESC UN3031 Chemistry of Climate
- EESC GR4926 Intro to Chemical Oceanography

Mathematics

- MATH UN1201-1202 Calculus III & IV
- MATH UN2005 Intro to Mathematical Proofs
- MATH UN2015 Linear Algebra and Probability
- MATH UN2030 Ordinary Differential Equations
- MATH UN2500 Analysis and Optimization
- APMA E4008 Advanced Linear Algebra
- APMA E4300 Numerical Methods
- APMA E4901-4903 Junior & Senior Seminars in Applied Math

Physics

- PHYS UN1401 Intro to Mechanics and Thermodynamics
- EESC GR4925 Intro to Physical Oceanography

Languages and Skills

Natural	Chinese (native), English (fluent), German (elementary)
Programming	Python, R
Markup	LaTeX, Markdown
Software	Google Suite, MS 365 (Microsoft Suite), iWork (Apple Suite)

Recommendations

Dr. Galen A. McKinley

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